Streaming Key Estimation with Abstention from Live Chord-Recognition Output

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Abstract. Most key-detection research estimates key from a complete score or recording, either as a whole-piece label or as an offline local analysis. We study a stricter interactive setting: a musician plays a MIDI keyboard, a chord recognizer emits a causal stream of symbolic chord hypotheses, and the key detector must update after each chord while abstaining when the evidence is insufficient. We define an evaluation protocol for streaming key estimation with abstention, including selective accuracy, coverage, stability, modulation lag, and piece-level paired statistics, and evaluate a family of causal detectors on four corpus-derived fixture sets. The final system is a filtered hidden Markov model (HMM) over profile-correlation emissions, with one constrained symbolic cue that redistributes evidence within same-tonic major/minor pairs. Three results are central. First, the memory dial is a timescale selector, not a noise-accuracy tradeoff: short memory fits local-key labels, while longer memory fits section-key labels. Second, same-tonic mode confusions can be reduced without disturbing tonic selection when symbolic chord-quality evidence is confined to parallel-key pairs. Third, adaptive temporal alternatives trade stability for faster key change detection and do not improve the selected section-key operating point. On held-out pop-song fixtures, the causal abstaining detector reaches statistical parity with standard offline music21 key finders that read the whole song before answering, while providing posterior probabilities, abstention, and stability measurements.

1 Introduction

An interactive music application identifies the chords a musician plays on a MIDI keyboard in real time. A natural next feature is a key indicator: the tonal center the player is in right now, inferred from what they have played. This is key finding, one of the oldest problems in music information retrieval, but the interactive setting differs from the literature's in three compounding ways.

First, the detector is **causal and streaming**. It only ever sees the past, and it must update after each chord, from someone noodling with finger rolls, pedal blur, and wrong notes. There is no second pass and no lookahead; the Viterbi decodings that dominate published HMM key-finding are unavailable by construction.

Second, the observations are **uncertain and symbolic at the chord level**. The input is not ground-truth notes or audio chroma but the output of a chord recognizer: ranked candidate identities with explanation costs, together with the voicing, timing, and hold duration of each committed chord. This setting is less studied than audio or score-based key detection, and it carries unusual assets (the detector knows each chord's root, quality, and bass) alongside unusual noise.

Third, **abstention is a first-class outcome**. A modal vamp has no single right answer, and asserting one confidently is worse than staying quiet. We therefore measure coverage and accuracy-on-claimed as an inseparable pair, along with stability, modulation lag, and time-to-first-claim, rather than accuracy alone.

The contributions are:

1. **A protocol for streaming key estimation with abstention**: metrics in the selective-prediction frame, frozen before tuning, with piece-level paired statistics for every adoption decision and a held-out evaluation performed exactly once.
2. **An explicit fixture-based data and reproducibility design**: chord streams are generated under a fixed neutral context, labels are stripped before detector calls, split files and result artifacts are versioned, and license-gated corpora are rebuilt locally rather than redistributed.
3. **A causal HMM specification for chord-recognition output**: filtered posteriors, profile-correlation emissions, a circle-of-fifths transition kernel, posterior-margin abstention, and a mass-preserving same-tonic mode cue.
4. **The timescale finding**: the emission-memory dial selects which granularity of key structure a causal detector reports. Dose-response curves on two annotation cultures run in opposite directions with no interior optimum (Figure 1), the crossover is significant on held-out data in both directions, and it reproduces within one corpus on identical performed input as a function of annotated segment length (Figure 2). Accuracy comparisons between key detectors are therefore label-timescale dependent.
5. **Measured negative results for added temporal and symbolic complexity**: recognizer-confidence weighting, progression rules under the HMM, relative-pair tilts, explicit-duration modeling, and Bayesian online changepoint detection each have documented outcomes.
6. **A held-out comparison**: the final causal detector reaches at least parity with offline whole-piece baselines, with point estimates ahead on every comparison, while abstaining and reporting streaming stability metrics.

2 Related work

Distributional key finding begins with the Krumhansl-Schmuckler probe-tone profiles [1] and their revisions [2], with corpus-trained profiles improving minor-mode behavior [3]. These methods estimate key from pitch-class distributions and remain important baselines because they are simple, reproducible, and musically interpretable. The selected detector in this paper keeps that interpretability: its emissions are profile correlations rather than learned sequence embeddings.

Temporal structure enters key and harmony estimation through Bayesian and hidden Markov treatments [4], [5]. The closest published analogue of our final detector is justkeydding [6], an HMM over profile emissions for global and local key. Neural symbolic-harmony systems also estimate local key as one head of a larger harmonic-analysis task, for example models using Roman numeral corpora and multitask architectures [7], [8]. Those systems demonstrate the value of local-key estimation, but their usual setting is offline score analysis. Our aim is narrower and more constrained: causal filtering over chord-recognition events with abstention and stability metrics.

Real-time tonal tracking has also been studied outside the profile/HMM family. Chew's Spiral Array and Center of Effect Generator frame key finding as movement in a geometric tonal space, tracking an evolving tonal center as notes accumulate [9]. Our setting differs in the observation layer: rather than raw pitch collections, the detector receives symbolic chord identities and candidate rankings produced by an upstream recognizer. That makes chord quality a usable signal, but it also means recognizer errors and candidate uncertainty are part of the input distribution.

Bayesian online changepoint detection [10] maintains a run-length posterior and is the natural adaptive-memory alternative to a fixed decay. Recent extensions model within-regime dynamics to suppress false alarms [11]. We build the constant-hazard version for the 24-key space and measure where its adaptivity helps and where it conflicts with the selected section-key operating point.

Evaluation practice follows the MIREX audio key task's weighted scoring [12] for near misses, applied to our own corpora (scores are not comparable to MIREX leaderboards, which are audio-based on different data). Ground truth comes from sources spanning two annotation cultures: When in Rome's analyst local keys and RomanText annotations [13], [14], the ASAP corpus of aligned performed piano MIDI [15], and Isophonics song-level keys via the ChoCo chord corpus [16], [17].

3 Task and evaluation protocol

Task. The input is a causal stream of chord events from the application's capture path: each event carries ranked chord candidates with explanation costs, the sounding voicing, the analysis-time tonality context, a timestamp, and a hold duration. After each event the detector outputs either a ranked list of (key, confidence) hypotheses or an explicit abstention. It never sees the future and is never re-run on edited history. The key space is the 24 major and minor keys over pitch classes; enharmonic spelling is a presentation concern.

Metrics. The top-ranked claim is scored. Accuracy-like metrics are reported in the selective-prediction frame: coverage (fraction of events with a claim) and accuracy on claimed events form an inseparable pair, and abstentions are never errors. Exact accuracy is complemented by the MIREX-weighted score (fifth 0.5, relative 0.3, parallel 0.2) [12]. Temporal behavior is measured by modulation matching and lag (an annotated key change is matched when the detector claims the new key before the next change; unmatched changes are censored counts, never averaged into lag), spurious switches per piece (a switch is spurious only when the annotation did not change and the new claim is not the annotated key), and time-to-first-claim. Ambiguity-labeled events (hand-authored fixtures only) accept abstention or any acceptable key. Selective-prediction behavior is reported as the coverage-accuracy curve swept over the abstention threshold, with the selected operating point marked (Figure 3). For probabilistic detectors, we also report top-label posterior reliability: events are binned by the posterior probability of the leading key (ten equal-width bins) and compared with exact correctness, alongside expected calibration error (ECE), negative log likelihood, and Brier score. This diagnostic is separate from abstention behavior: a detector can abstain informatively while still being overconfident in its posterior probabilities.

Statistics. Every adoption decision uses piece-level paired comparisons: Wilcoxon signed-rank tests with seeded-bootstrap 95% confidence intervals on the per-piece mean delta. Events within a piece are strongly dependent and a few long pieces dominate pools, so the piece is the unit of analysis and pooled event accuracy is never decisive. The signed-rank test assumes no distributional form for these small delta samples. Development p-values served model selection, uncorrected for multiplicity; confirmatory weight rests on the pre-declared held-out evaluation, performed exactly once (Section 9).

Protocol discipline. The protocol was frozen before detector tuning, with changes recorded as dated amendments. Development/test splits were frozen per corpus before the first experiment on it (by piece, and by composer where the corpus allows); all tuning, ablation, and model selection ran on development splits; the test splits were evaluated exactly once, with the result set declared in advance (Section 9). The evaluation harness structurally strips labels before events reach a detector.

4 Data and reproducibility

Evaluation uses **fixtures**: versioned event streams containing the chord recognizer's candidate rankings, the observed voicings, timing metadata, and a parallel label stream read only by the scorer. Fixtures are generated under a fixed neutral analysis

corpus	input	labels	pieces	events
When in Rome [13]	quantized scores	analyst local keys (local-key)	77	5,207
ASAP [15]	performed piano MIDI	key signatures (section-key, mode-unknown)	60	19,546
Isophonics [16], [17]	synthesized voicings	song keys (section-key, mode-resolved)	224	19,062
ASAP-WiR overlap	performed piano MIDI	analyst local keys (local-key)	36	10,395

Table 1: Evaluation corpora. Development/test splits are frozen for the first three; the overlap corpus is evaluation-only (every configuration run on it was committed in advance).

context so tonality-gated ranking rules cannot leak ground truth into the observations. Results are comparable only within a fixture version, because fixture contents embed a specific version of the chord-recognition engine.

Four fixture sets span two annotation cultures and three input conditions (Table 1). The first three have frozen development/test splits. The ASAP-WiR overlap is evaluation-only: every configuration run against it was declared in advance because the set exists specifically to test whether the timescale effect survives on identical performed input.

The overlap corpus transfers When in Rome analyst keys onto ASAP performances of the same Beethoven sonata movements through the performance-to-score downbeat alignment, giving real tonic-and-mode ground truth on performed input. A small hand-authored pop/jazz suite (12-bar blues, modal vamps, deliberately ambiguous loops) serves as a per-fixture behavioral regression battery outside all pooled statistics. Performed corpora are replayed through the application's actual capture code (pedal-aware sounding sets, debounce, minimum-duration commit), so detectors see the events they would see live.

The repository contains the open fixtures, split files, evaluation harness, paired-comparison script, dated experiment logs, and the committed artifacts of the single held-out evaluation. License-gated corpora are not committed: their extractors refuse to write inside the research directory and must be rerun locally from pinned upstream checkouts. This means the paper's reported numbers are auditable from committed reports and commands, while redistribution remains bounded by each source corpus's terms.

5 Model family and selected detector

The detector family was deliberately interpretable. The profile-correlation floor maintains a duration-weighted, exponentially decaying pitch-class histogram and ranks all 24 rotations of a major/minor profile pair by Pearson correlation, using the corpus-trained profile values of [3] (whose original algorithm pairs them with Euclidean distance). Two symbolic additions were also built: a weighted-evidence layer scoring key-relative chord functions (diatonic membership, dominant and leading-tone function), and a progression layer voting on cadential transition patterns. These layers form a three-way hybrid (correlation base plus small functional and progression blend terms) that beat the floor on early local-key development runs (+0.096 exact per piece, $p = 0.003$), but later ablations show that the selected section-key configuration should remove them. Thus, except for the selected same-tonic mode cue below, chord-identity features proved useful mainly for the local-key version of the task; the section-key configuration is deliberately closer to a profile model.

The selected detector is a causal HMM over the 24 major and minor keys. It keeps a filtered posterior and updates it by the forward algorithm only; no Viterbi decoding or future context is used. At each event, the update is:

```
for each pitch class p:
  histogram[p] =
    sum over prior events i of:
      duration_i
      * 2^(-age_i / half_life)
      * indicator(p in voicing_i)

for each candidate key:
  profile_score[key] =
    correlation(histogram, key_profile[key])

emission = softmax(profile_score / temperature)
prior =
  transpose(transition) * previous_posterior
posterior = normalize(emission * prior)
```

Multiplication in the last pseudocode line is elementwise; the transition matrix supplies persistence over hidden keys, and the normalized posterior is the filtered belief after the current event. In words, the profile scores are converted into an event-level likelihood over keys, the previous belief is advanced through the transition model to form a prior, and the two are combined and renormalized.

In the selected configuration, `key_profile` is the Albrecht-Shanahan major/minor profile set. Each event's contribution to the histogram is multiplied by its duration, and `half_life = 30 s` means that contribution halves after 30 seconds. The softmax temperature is 0.25, sharpening the profile-correlation scores into a more selective likelihood. The transition matrix assigns 0.9 probability to remaining in the same key; the remaining mass is distributed over other keys with decay by circle-of-fifths signature distance. At each event, the detector reports the posterior's top key and its posterior probability only when the margin between the top two posterior probabilities is at least 0.3; otherwise it abstains.

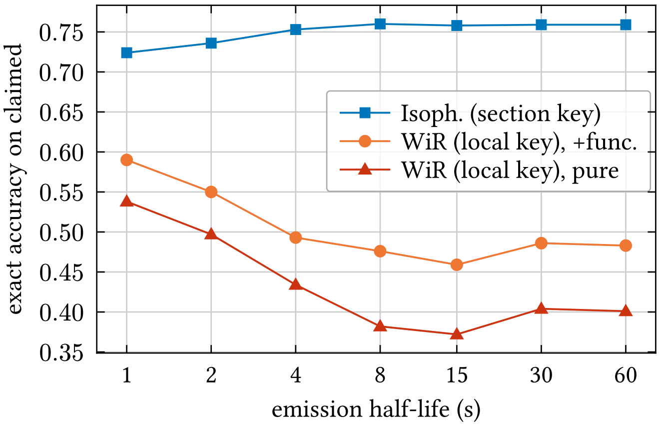

Figure 1: Emission-memory dose-response on the two development annotation scales. The curves run in opposite directions with no interior optimum: the dial selects the reported timescale. Coverage rises with memory on both scales (0.77 to 0.93 on Isophonics); the section-key plateau spans 8-60 s. "Pure" means profile-correlation emissions only; "+func." adds the early chord-function evidence term.

Two modeling constraints matter for interpreting the results. First, the HMM's state persistence and the emission scorer's context window are distinct: the decayed histogram defines the current observation, while the HMM supplies temporal persistence over hidden keys. Reintroducing history both as an emission blend and as state persistence double-counts evidence. Second, the emission memory controls how local each observation is. The next section shows that this setting selects between annotation targets.

6 The timescale finding

Key annotations come in two cultures. Analyst local keys (When in Rome) mark every tonicization, about seven key regions per piece at a few events each. Song keys and key signatures (Isophonics, ASAP) name the section key and absorb excursions. We call these **local-key** and **section-key** annotations: the former reward immediate response to local key assertions, while the latter reward stability around the longer musical home.

Sweeping the emission-memory half-life across 1 to 60 seconds on both development annotation scales gives the dose-response curves of Figure 1. On the local-key labels, accuracy falls essentially monotonically with memory and modulation matching halves; on the section-key labels, accuracy climbs to a broad plateau from 8 s outward while coverage and stability keep improving (spurious p90 falls from 7 to 1). The relevant result is the crossover between annotation targets, not a single best smoothing value.

Three controls close the alternative explanations. The crossover survives noise: on the evaluation-only overlap corpus, identical performed recordings scored against analyst keys prefer short memory (0.59 vs 0.47 exact), matching the clean-score result. It reproduces **within** one corpus as a function of segment length (Figure 2). And it holds on held-out data with significance in both directions (Section 9).

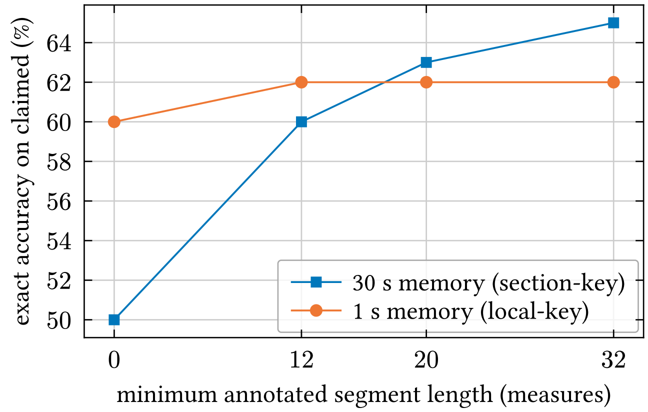

Figure 2: The crossover within a single corpus on identical performed input (ASAP-WiR overlap): filtering to longer analyst key segments, the long-memory configuration climbs and overtakes the short-memory one at 20-measure segments, which stays flat. The same recordings, sliced by label granularity, reorder the two configurations.

The application conclusion is a choice, not a claim that one timescale is universally better: a glanceable key indicator should report the section, so the selected configuration uses 30 s memory and deliberately absorbs brief tonicizations. The research conclusion is that cross-paper accuracy comparisons are under-specified without the label timescale: the same detector moves by tens of points when the annotation scale changes.

Ablations. Two full 16-cell factorials (functional blend, progression blend, duration weighting, recognizer-confidence weighting) ran on each annotation scale. The functional blend adds key-relative chord-function points to the profile score; the progression blend adds cadential-transition points; duration weighting uses hold duration instead of one vote per event; and recognizer-confidence weighting downweights near-tied chord recognitions using the best-to-second explanation-cost gap. The functional blend flips sign with the annotation scale: removing it is a significant paired win for section-key annotations, while for local-key annotations it is load-bearing (+0.061 exact, CI95 [+0.019, +0.108], $p = 0.010$). This is direct evidence that the functional rules primarily detect brief excursions. The progression blend's earlier value was architecture-specific and washes out under the HMM. Duration weighting helps on both scales (+0.02 to +0.05). Recognizer-confidence weighting is inert in five out of five paired tests, a useful negative result because exploiting recognizer confidence was a natural hypothesis for this setting. The selected emissions are therefore pure profile correlation, and a long-standing behavioral failure (a 12-bar blues misread as its subdominant) vanished with the deleted rules rather than through any added mechanism.

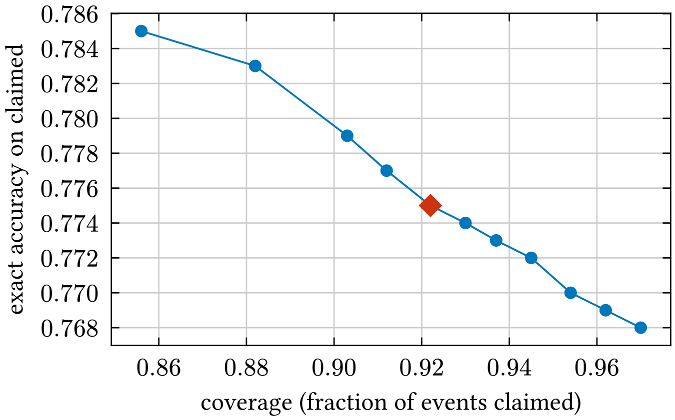

Figure 3: Selective-prediction behavior: the coverage-accuracy curve of the selected configuration on the Isophonics development split, swept over the posterior-margin floor (0 to 0.6). Moving left raises the margin required to speak, so coverage falls; moving up means the remaining claims are more often correct. The marked point is the selected operating point (floor 0.3). The curve rises monotonically as the detector grows choosier: abstentions are informative, not random.

7 Mode disambiguation

On mode-resolved corpora the residual errors sort into MIREX classes: fifth-neighbor confusions, relative-pair confusions (shared signature), and parallel-pair confusions (shared tonic). Mode errors, the most user-visible class since an indicator displays major or minor, were about 10% of claims on both annotation scales once the timescale trade was filtered out.

The useful constraint is mass preservation within a musically related pair: when evidence is redistributed only between two hypotheses sharing an invariant, the cue can help decide between those hypotheses without adding evidence to unrelated keys.

Parallel pairs (selected). Parallel keys share the tonic pitch class, and practical minor borrows the raised sixth and seventh, so the discriminating evidence collapses to roughly one pitch class and is easily outvoted in a 12-dimensional correlation. But our observations are symbolic: when the played chord is rooted on a candidate tonic with a home quality, its major or minor quality is the most direct mode cue available. Applied as a log-odds tilt within the parallel pair after the emission softmax, and then rescaled so the pair's emission sum is unchanged, the cue cannot move tonic evidence at all. This prevents the failure mode observed for less constrained tonic-bonus and functional-rule variants, which could fight modulation tracking. Strength 2 was selected after a 0.25-to-4 sweep with a 2-to-4 plateau: paired exact wins on both annotation scales (Isophonics +0.016, p = 0.030; When in Rome +0.030, p = 0.029), parallel confusion roughly halved everywhere measured (4% to 2% of claims on Isophonics), and no stability cost on the section-key development set.

Relative pairs (measured negative). The same pattern generalizes with the key signature as the conserved quantity. But the evidence is structurally weaker: the rival's tonic chord is ordinary diatonic harmony (the vi chord), not rare mode mixture, so isolated chord quality fires constantly for the wrong twin. Two sharpened cues were built and swept: a bass-gated tilt (fires only when the chord's root is also its bass) and a cadential-bigram tilt (a dominant-quality chord resolving down a fifth onto a home-quality chord). The cadence cue is inert, echoing the progression ablation. The bass cue works mechanically (relative confusion 6% to 4% of claims) but misses paired significance on the primary scale (+0.007, p = 0.055), trends negative on the other (-0.009, p = 0.056), and doubles spurious p90; unlike the parallel tilt's broad plateau it turns harmful at strength 2, the signature of a weak signal. Not adopted; retained in code as a measured negative.

Fifth pairs (not adopted). Fifth neighbors conserve neither tonic nor signature, and every key has two of them, so there is no pair to redistribute within; discriminating a key from its dominant **is** the modulation problem, and any chord-quality nudge between fifth neighbors is the already-refuted tonic bonus reborn. Empirically the residual fifth errors also lack an exploitable shape, splitting nearly evenly between the dominant and subdominant sides (57/43 and 56/44 on the two mode-resolved corpora).

The pattern explains an adoption, a failure, and a non-adoption: the tilt's value tracks the strength and exclusivity of the invariant it preserves (tonic: strong; signature: weak; none: unavailable).

8 Adaptive temporal models

Explicit-duration modeling (HSMM). The HMM's self-transition implies geometric key-dwell times; an HSMM's contribution is a richer dwell family. Before building one we measured whether the system responds to the dwell parameter it would refine: sweeping self-transition from 0.7 to 0.98 (implied mean dwell roughly 3 to 50 events) moves every metric along the coverage-accuracy curve rather than off it (Isophonics exact within 0.008, spurious p90 pinned at 1). This probe does not prove that explicit-duration modeling is never useful for key estimation, but it shows little headroom for the selected operating point.

Bayesian online changepoint detection, built and measured. The probe addresses the changepoint prior (constant-hazard BOCPD implies the same geometric dwell) but not BOCPD's distinct contribution, the run-length-adaptive evidence window: pooling evidence back to the inferred section start instead of through a fixed 30 s decay. We built the constant-hazard detector for the 24-key space [10] with emission conventions identical to the selected HMM (including the mode tilt), so differences isolate the window (Table 2).

config	cov	exact	modulations	spur med/p90
HMM, selected	0.92	0.775	94/192	0/1
BOCPD h=1/200	0.88	0.715	161/192	5/14
BOCPD T=0.5	0.89	0.765	135/192	2/7
BOCPD T=1.0	0.90	0.769	115/192	0/4

Table 2: BOCPD versus the selected HMM on the section-key development set (Isophonics dev). Here `h` is the constant changepoint hazard and `T` is emission temperature. The adaptive window improves modulation detection but no tuning recovers the selected stability point.

The adaptive window delivers its promise: modulation matching jumps by 70%, because evidence resets at inferred section starts. But the same reactivity breaks section-key stability, and no tuning recovers the selected operating point: softening the per-event evidence walks BOCPD back along the frontier without reaching the HMM (best cell: exact wash, $p = 0.51$, spurious p90 still 4 versus 1). One finding worth keeping: at matched reactivity BOCPD dominates the HMM's fast settings (0.765 exact at 135 modulations versus 0.736 at 141 for the HMM at a 2 s half-life), so adaptive windowing has value for local-key tracking; it was not selected for the section-key task studied here. The false alarms are within-section harmonic movements, exactly where BOCPD's independent-given-key assumption is weakest. Autoregressive BOCPD extensions target analogous within-regime dependence in continuous series [11], but those Gaussian remedies do not transfer directly to categorical chord emissions, and our domain-specific substitute (the progression layer) is measured inert.

9 Held-out evaluation

The test splits were evaluated exactly once, with the result set declared before any run: the selected configuration on all three splits, the local-key reference on the two mode-resolved splits, the music21 baselines on Isophonics with paired and matched-coverage comparisons, and a descriptive mode-confusion breakdown. The complete artifacts are committed with the project.

split	pieces	cov	exact	mods	spur
Isophonics	41	0.88	0.732	10/22	0/3
WiR	18	0.81	0.587	39/115	0/1
ASAP	10	0.83	0.683	-	0/0

Table 3: The selected configuration on the three held-out splits. `mods` is matched annotated modulations; `spur` is median/p90 spurious switches. WiR is When in Rome; ASAP is scored with acceptable keys because labels are mode-unknown key signatures.

Generalization is clean (Table 3): Isophonics dips modestly from development (0.775 to 0.732), When in Rome comes in far above it (0.434 to 0.587, easier held-out pieces), and stability holds everywhere; the differences run in both directions, not the uniform degradation a tuned-to-development configuration would show.

The held-out crossover is significant in both directions. The short-memory local-key configuration is better when judged against local-key labels (0.649 vs 0.587 exact, paired +0.062, CI95 [+0.004, +0.121], $p = 0.047$). The selected section-key configuration is better when judged against section-key labels (0.732 vs 0.556, paired +0.175, CI95 [+0.040, +0.315], $p = 0.039$), while also avoiding the local-key configuration's spurious switches (0/3 vs 5/11 med/p90).

External comparison. Table 4 compares against music21′s offline whole-piece analyzers on the held-out songs.

system	coverage	exact	MIREX
this work (causal, abstaining)	0.88	0.732	0.782
Temperley-Kostka-Payne	1.00	0.637	0.740
Krumhansl-Schmuckler	1.00	0.624	0.726
Aarden-Essen	1.00	0.558	0.690

Table 4: Held-out Isophonics test split: the causal, abstaining detector against offline whole-piece baselines.

The causal detector's point estimate leads every offline analyzer, including at matched coverage (Krumhansl-Schmuckler restricted to our claimed events: 0.63). The paired test is a wash (+0.108, CI95 [−0.008, +0.228], $p = 0.25$), so the defensible claim is **at least parity under a more constrained setting**: the baselines read each entire song before answering; this detector never sees the future, abstains on the ambiguous 12%, and reports posterior probabilities and streaming stability metrics offline systems do not have.

The descriptive confusion mix is: exact 72%, fifth 7%, relative 5%, parallel 2%, other 14%. This pooled exact rate differs from the per-piece mean in Table 3; the parallel row matches the mode tilt's development effect exactly.

10 Limitations

The label timescale is a construct of the annotation culture, and our central finding is precisely that results are relative to it; we mitigate by reporting both annotation scales, but neither scale is "the truth."

Residual error contains genuine ambiguity (relative twins share every pitch class; mode-mixture passages and dominant prolongations can reasonably be notated different ways), so ceilings are below 1.0 and unknown.

The recognizer sits inside the loop: fixtures embed its rankings, so recognizer changes can change detector results and require regenerating fixtures before comparison.

Corpus licensing constrains redistribution (two corpora are noncommercial- or research-gated; we commit derived facts, splits, and evaluation artifacts, not the gated fixtures).

Test splits are small (10 to 41 pieces), which the paired statistics respect but cannot cure. Stronger neural local-key systems are offline and score-hungry; we compare only against baselines runnable on our fixtures.

Raw posteriors are overconfident: on claimed exact-labeled development events, mean top posterior is 0.929 while exact accuracy is 0.772 (expected calibration error, ECE, 0.157). This does not affect ranking, abstention, or paired accuracy claims. For display only, post-hoc temperature scaling [18] fit on the development split (T = 1.55, negative log-likelihood argmin) reduces held-out claimed-event ECE from 0.192 to 0.041 without changing any claim (byte-identical to the committed held-out artifacts). Calibration remains task-relative: against local-key labels, the same display remains overconfident.

11 Conclusion

This paper defines and evaluates streaming key estimation with abstention from live chord-recognition output. Under a frozen protocol with paired statistics and a held-out evaluation performed exactly once, a causal HMM over profile-correlation emissions, augmented by one mass-preserving same-tonic mode cue, reports section keys on live-style chord streams at statistical parity with offline whole-piece baselines. The route to that detector was largely measurement-guided simplification: functional and progression rules were useful for local-key tracking but harmful or unnecessary for the selected section-key setting; recognizer-confidence weighting was inert; and adaptive temporal models traded stability for reactivity. The central methodological lesson is that key-detection accuracy is label-timescale dependent. A detector can appear better or worse depending on whether the reference labels ask for brief tonicizations or section key, so the annotation scale is part of the task definition rather than a detail of the dataset.

Reproducibility: protocol, logs, corpus pins, splits, harness, and held-out evaluation artifacts live in the WhatChord repository under github.com/EarthmanMuons/whatchord/tree/main/research/whatkey . Every number traces to a dated log entry and a report with its generating command.

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